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Research Paper

Rapid Detection of *Salmonella* Typhimurium Microcolonies Using Phase-Contrast Imaging and Deep Learning

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ABSTRACT

Early detection of foodborne pathogens is critical for improving food safety and preventing contamination-related outbreaks. Conventional methods for detecting *Salmonella* spp. are reliable but require extended incubation times and labor-intensive procedures, limiting their use for rapid screening. In this study, we developed a rapid and automated detection framework that integrates phase-contrast microscopy with a YOLOv8x-based deep learning model to identify *Salmonella enterica* serovar Typhimurium at the microcolony stage. A dataset of 5,000 microscopy images, including pure and mixed cultures with *Escherichia coli* K-12 under both clean and food-matrix (onion) conditions, was collected across multiple time points (30 min to 4 h). The model achieved a precision of 0.891, recall of 0.867, and mAP@0.50 of 0.898. Reliable detection was achieved as early as 2 h of incubation, significantly reducing the time required compared to conventional culture-based methods. The model also demonstrated robust performance in mixed cultures, effectively distinguishing between morphologically similar bacterial species. In addition, instance-level analysis enabled quantitative characterization of microcolony growth dynamics over time. This approach provides a rapid, accurate, and cost-effective strategy for early-stage pathogen detection and has strong potential for real-time food safety monitoring and microbial diagnostics.

Detecting foodborne pathogens early is essential to ensure food safety and enable timely treatment of infections. Foodborne diseases represent a major global health challenge, creating significant economic and societal burdens, as pathogenic microorganisms contaminate food products and lead to numerous illnesses and outbreaks among consumers (Vidyadharani et al., 2022). Estimates suggest that these illnesses result in annual losses approaching US $90 billion (Scharff, 2020). Consequently, the advancement of innovative, high-tech approaches for rapid, precise, and less labor-intensive microbial detection is essential for enhancing global food safety. These emerging solutions are increasingly meeting the demand for detection methods that are both faster and more sensitive than conventional techniques (Cho & Ku, 2017).

Salmonella spp. are Gram-negative bacteria encompassing more than 2,500 recognized serotypes. Among them, nontyphoidal *Salmonella* (NTS) is one of the leading causes of bacterial foodborne illness globally and poses a persistent public health threat (Kumar et al., 2025). Infection can result in self-limiting gastroenteritis, invasive bacteremia, and severe focal complications, particularly in vulnerable populations (Roth et al., 2018). The continuing emergence of multidrug-resistant strains further complicates treatment and control efforts (Chen et al., 2013). In the United States, NTS infections account for a substantial proportion of foodborne-related hospitalizations and deaths (Hoffmann et al., 2012). The Centers for Disease Control and Prevention (CDC) estimates that NTS causes approximately 1.2 million illnesses and about 450 deaths each year (CDC, 2023), representing more than one million of the estimated 3.6 million cases of bacterial foodborne disease nationwide (Scallan et al., 2011). Beyond the public health impact, contamination of poultry and pork products with *Salmonella* results in considerable economic losses, estimated at approximately $2.8 billion and $1.9 billion annually, respectively (Scharff, 2020).

Contamination is not limited to animal-derived foods. Fresh produce can become contaminated at multiple points along the production chain, including during cultivation through contaminated irrigation water or manure, as well as during harvesting, processing, transportation, or packaging (McCormic et al., 2022). This risk is reflected in recent multistate outbreaks linked to contaminated onions

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in 2020, 2021, and 2023, highlighting the ongoing vulnerability of produce supply chains to *Salmonella* contamination (CDC, 2023). These events emphasize the necessity for highly sensitive and reliable detection methods capable of identifying pathogens in complex food matrices before widespread distribution occurs.

Current detection approaches for foodborne pathogens include culture-based and nonculture-based methods (Awang et al., 2021). Although culture-based methods are well established and highly reliable, they typically require enrichment procedures and extended incubation periods, often delaying results for several days. Nonculture-based methods for *Salmonella* detection include immunological, molecular, and biosensor approaches that are faster than traditional methods (Wang et al., 2021). Molecular techniques are highly accurate, while immunological tests are simpler but less reliable. Molecular assays such as PCR can enhance detection sensitivity and specificity; however, the PCR workflow itself is demanding, requiring costly equipment as well as trained personnel to perform and interpret the results (Park et al., 2014). Advanced tools like biosensors offer rapid detection but are often limited by cost and technical demands (Riu and Giussani, 2020). Importantly, most conventional methods detect pathogens only after substantial bacterial proliferation has occurred, reducing the opportunity for early intervention and increasing the likelihood that contaminated products will enter the food supply.

Recent progress in artificial intelligence (AI) and computer vision has opened new possibilities for automated microbial analysis (Bonah et al., 2019). Nevertheless, the application of these technologies to detect bacteria at very early stages of growth, particularly at the microcolony stage, remains limited. Consequently, there is a critical need for rapid, nondestructive techniques capable of detecting *Salmonella* before visible colony formation without lengthy enrichment or complex processing steps. We hypothesize that the integration of phase-contrast microscopy with a YOLOv8x-based deep learning model enables rapid and accurate detection of *Salmonella* Typhimurium microcolonies at early incubation stages, even in complex sample backgrounds.

To address this unmet need, this study presents a novel, rapid detection framework that combines phase-contrast microscopic imaging with artificial intelligence-driven real-time object detection. Using convolutional neural network (CNN) models such as YOLOv8x, the system demonstrates the potential to identify *Salmonella* microcolonies shortly after cultivation. This integrated approach offers a rapid, accurate, and cost-effective alternative to conventional methods and has the potential to significantly improve food safety monitoring by enabling earlier detection of contamination events and more timely preventive actions.

Materials and methods

Bacterial strains and culture conditions. Two Gram-negative bacterial strains, Salmonella enterica serovar Typhimurium and nonpathogenic E. coli K-12, were used in this study. The *S.* Typhimurium strain was obtained from the Food Microbiology Laboratory at North Carolina A&T State University. Lyophilized E. coli K-12 cells were purchased from Sigma-Aldrich (St. Louis, MO, USA). Both strains were cultured in tryptic soy broth (TSB; dehydrated culture medium, Sigma-Aldrich, India). An *E. coli* K-12 strain was selected as a representative organism because *E. coli* and *S.* Typhimurium are phylogenetically related enteric bacteria that exhibit comparable colony morphologies and biofilm-associated multicellular behaviors.

For long-term storage, equal volumes of bacterial culture and sterile 50% (v/v) glycerol solution were mixed thoroughly and stored at −80 °C. Prior to each experiment, frozen stocks were streaked onto tryptic soy agar plates (TSA; Sigma-Aldrich) and incubated at 37 °C for 18–24 h. A single colony from each plate was then inoculated into 10 mL of TSB and incubated overnight at 37 °C with shaking at 175 rpm.

The resulting overnight cultures were diluted with sterile Dulbecco's phosphate-buffered saline (DPBS; Sigma-Aldrich, St. Louis, MO, USA) to obtain a final concentration of approximately 10^6 CFU/mL for microcolony cultivation, as confirmed by optical density calibration.

Microscopic dataset collection. Glass Petri dishes (50 mm; Eisco™, Fisher Scientific, USA) were sterilized by overnight dry-heat sterilization. Five circles, each approximately 5 mm in diameter, were marked on the outer bottom surface of each dish using a permanent marker to guide the placement of aliquots at the center of each circle. Soft *Salmonella*–Shigella (SS) agar (Sigma-Aldrich) was used for imaging *Salmonella* Typhimurium monocultures, with and without onion samples. Soft HiCrome™ Chromogenic Coliform Agar (CCA; HiMedia) was used for imaging *E. coli* K-12 monocultures, with and without onion samples. For experiments involving mixed cultures of *Salmonella* Typhimurium and *E. coli* K-12, a soft layer of *Salmonella* ChromoSelect Agar, improved for *Salmonella* and *E. coli* (Sigma-Aldrich), was used. This selective chromogenic medium is specifically formulated to support the simultaneous growth and differentiation of both organisms while suppressing competing background microorganisms.

A very thin layer (~1 mL) of the appropriate selective medium was aseptically transferred into labeled Petri dishes. To improve detection accuracy and minimize interference from sample debris, images of microcolonies grown without onion tissue were first captured. Subsequently, 2.5 μL of bacterial culture, with or without onion samples, was placed at the center of each marked circle. Onion samples were included to simulate a realistic food matrix and evaluate model performance under complex background conditions relevant to food contamination scenarios.

All inoculated plates were incubated at 37 °C for up to 4 h. Microcolony images were captured, saved, and labeled at 30-minute intervals using an IX73 inverted microscope (Evident Scientific, Inc., USA). Images were acquired with a digital color camera (Olympus LC35, USA) operated via cellSens software, producing images at a resolution of 2,160 × 1,620 pixels from a 3.5-megapixel CMOS sensor. Air objectives of 60x (PH2, NA 0.7, WD 1.5–2.2 mm) and 10x plan achromat (PH1, NA 0.25, WD 10.6 mm) were used with a TH4-100 transmitted light source (Olympus, USA). This imaging approach was implemented to reflect natural biological variation in microcolony size, morphology, and background complexity.

Raw dataset processing. Prior to training and testing the YOLOv8x model, all images were preprocessed using random flipping and normalization-based data augmentation, which accelerates training convergence and enhances the model's generalization capability (Ioffe & Szegedy, 2015).

A comprehensive dataset of 5,000 microscopy images was collected, encompassing pure cultures, mixed cultures, and background samples. Images were acquired under two conditions: with onion and without onion treatment, with equal representation across both conditions.

The dataset included images of *S.* Typhimurium, *E. coli* K-12, mixed bacterial cultures, and background samples without bacterial cells. Mixed-culture images were included both in training and evaluation to assess model generalization under multispecies conditions. All images were collected across eight time points (30 min, 1 h, 1.5 h, 2 h, 2.5 h, 3 h, 3.5 h, and 4 h) to capture the complete progression of bacterial growth from individual cells to mature microcolonies. Early-stage images (30 min–1.5 h) were included to represent the full range of morphological development during training, whereas the 2-h time point was identified as the earliest stage at which microcolonies consistently exhibited sufficient features for reliable detection.

The distribution of images across experimental conditions is summarized in Table 1.

Table 1
Composition of the microscopy image dataset

Category	With Onion	Without Onion	Total Images
S. Typhimurium (pure culture)	800	800	1600
E. coli K-12 (pure culture)	800	800	1600
Mixed cultures	800	800	1600
Background (no bacteria)	100	100	200
Total	2500	2500	5000

Image annotation and YOLOv8x model configuration. After capturing the images, we annotated the samples in two complementary formats, rectangular bounding boxes and polygonal segmentation masks, using the Computer Vision Annotation Tool (CVAT). This dual annotation approach enables both object detection tasks and detailed spatial analysis. The finalized annotated dataset was divided into 70% for training, 10% for validation, and 20% for testing.

The YOLOv8x model was implemented using the Ultralytics YOLOv8 framework in Python with PyTorch as the backend. The baseline model is composed of a CSP-Darknet backbone for extracting features, a PAN-FPN neck for aggregating multiscale information, and a detection head functioning across three spatial resolutions. YOLOv8x was selected as a single-stage detector, in which localization and classification are performed in a single forward pass rather than through sequential region proposal and refinement as in two-stage detectors such as Faster R-CNN. Single-stage architectures provide substantially higher inference throughput (Carranza-García et al., 2020), which is a practical requirement for the intended application, where large numbers of microscopy fields must be screened rapidly. We note that this represents a trade-off rather than a strict advantage, as two-stage detectors can achieve higher localization accuracy in visually complex scenes; for high-throughput screening, inference efficiency was prioritized. The three-scale detection head provides multiscale representation relevant to the considerable variation in microcolony size across the incubation window examined here, and the largest YOLOv8 variant (approximately 68 million parameters) was selected to maximize representational capacity. The YOLO family has established precedent in this specific imaging modality and biological application: Ma et al. (2023) demonstrated detection and species-level discrimination of bacterial microcolonies under phase-contrast microscopy after 3 h cultivation, and YOLOv8-derived architectures have subsequently been adapted for small-colony detection (Wang et al., 2025). The present study was designed as a proof-of-concept establishing the imaging and detection workflow rather than as a benchmarking study, and a single well-characterized architecture was therefore used throughout.

All models were trained under consistent experimental conditions using a fixed input image resolution of 640 × 640 pixels. Training was conducted using the default hyperparameters of the YOLOv8 framework without additional tuning to ensure reproducibility and consistency.

Model performance was evaluated using precision, recall, and mean average precision at an intersection-over-union threshold of 0.50 (mAP@0.50), which are standard metrics for object detection tasks and are particularly relevant for safety–critical applications such as foodborne pathogen detection.

Detection of *S.* Typhimurium in onion samples. Fresh white onion samples ($n = 20$) were purchased from local grocery stores in Carbondale, IL, USA, at a rate of two samples per week and transported to the laboratory under ambient conditions for immediate analysis. For each sample, 15 g of onion tissue was aseptically excised and inoculated by immersion for 5 min in freshly prepared, diluted tryptic soy broth (TSB) containing approximately 10^6 CFU/mL of *S.* Typhimurium to simulate surface contamination. The inoculated tissue was then transferred to a sterile 18-oz sampling bag (Weber Scientific, USA) containing 40 mL of phosphate-buffered saline (PBS) and homogenized using a laboratory blender (Seward, UK) at 230 rpm for 30 s to ensure uniform bacterial distribution throughout the sample.

A 2.5 μL aliquot of each homogenized sample was aseptically placed at the center of each of the five marked circles on the prepared agar plates. Three replicate plates were prepared for each sample and incubated at 37 °C for up to 4 h. Microcolony images were captured every 30 min, yielding a total of 1,600 images of *S.* Typhimurium microcolonies, including 100 images without onion and 100 images with onion at each time point.

Conventional plating methods were performed in parallel using the same selective agar medium. Colony-forming units (CFUs) were counted after incubation at 37 °C for 18–24 h. All experiments were conducted in triplicate across three independent days.

Detection and identification of *S.* Typhimurium when mixed with *E. coli* K-12. To assess the performance of the trained YOLOv8x model, E. coli K-12 was selected as a second strain due to its similar rod-shaped morphology to *S.* Typhimurium (Moxley, 2022). An overnight culture of both *S.* Typhimurium and E. coli K-12 was prepared and subsequently diluted 1:1,000 in PBS and mixed at a volumetric ratio of 8:2 (*S.* Typhimurium: *E. coli* K-12). Specifically, 8 mL of diluted overnight *S.* Typhimurium culture was aseptically combined with 2 mL of diluted overnight *E. coli* K-12 culture to obtain a total volume of 10 mL. Five drops, each approximately 2.5 μL, were dispensed onto soft chromogenic agar plates and incubated at 37 °C for up to 4 h. Microcolony images were captured every 30 min at eight time points (30 min, 1 h, 1.5 h, 2 h, 2.5 h, 3 h, 3.5 h, and 4 h), yielding a total of 1,600 images. In parallel, conventional plating methods were performed to quantify the proportion of each bacterial strain by enumerating colonies from the respective overnight cultures.

Quantitative measurement of microcolony size using instance segmentation. Microcolony size was quantified using instance-level segmentation masks generated from annotated microscopy images, following the methodology described in our previous time-stamped imaging study (Koirala et al., 2025). The area of each colony was calculated by counting the number of pixels within the segmented region and converting this value to μm^2 using microscope calibration (0.125 μm per pixel). This approach provides an objective, accurate, and reproducible measurement of colony size across different time points. All measurements were performed consistently across the dataset using the same calibration parameters.

Statistical analysis. All experiments were performed in triplicate, and data are presented as mean ± standard deviation. Statistical analyses were conducted to evaluate differences among groups, including incubation times and treatment conditions (with and without onion). One-way analysis of variance (ANOVA) was used, followed by Tukey's honestly significant difference (HSD) test for posthoc comparisons when significant differences were observed. Before analysis, data were assessed for normality and homogeneity of variance. A p value < 0.05 was considered statistically significant. All statistical analyses were performed using Microsoft Excel (Microsoft Corp., Redmond, WA, USA).

Results

Microcolony development and time-resolved imaging. The growth dynamics of *S.* Typhimurium microcolonies were monitored using IX73 phase-contrast microscopy (Fig. 1). Individual bacterial cells underwent elongation and division during the early incubation period (1–2 h), forming small aggregates (Su et al., 2012). Distinct microcolonies became visible between 2 and 3 h and expanded into larger, dense structures by 3.5–4 h. A statistically significant increase in microcolony size was observed across incubation time points ($p < 0.05$, one-way ANOVA). Posthoc analysis using Tukey's HSD test confirmed significant differences between early (≤2 h) and later (≥3 h) incubation stages, as well as between successive time points. A transi-

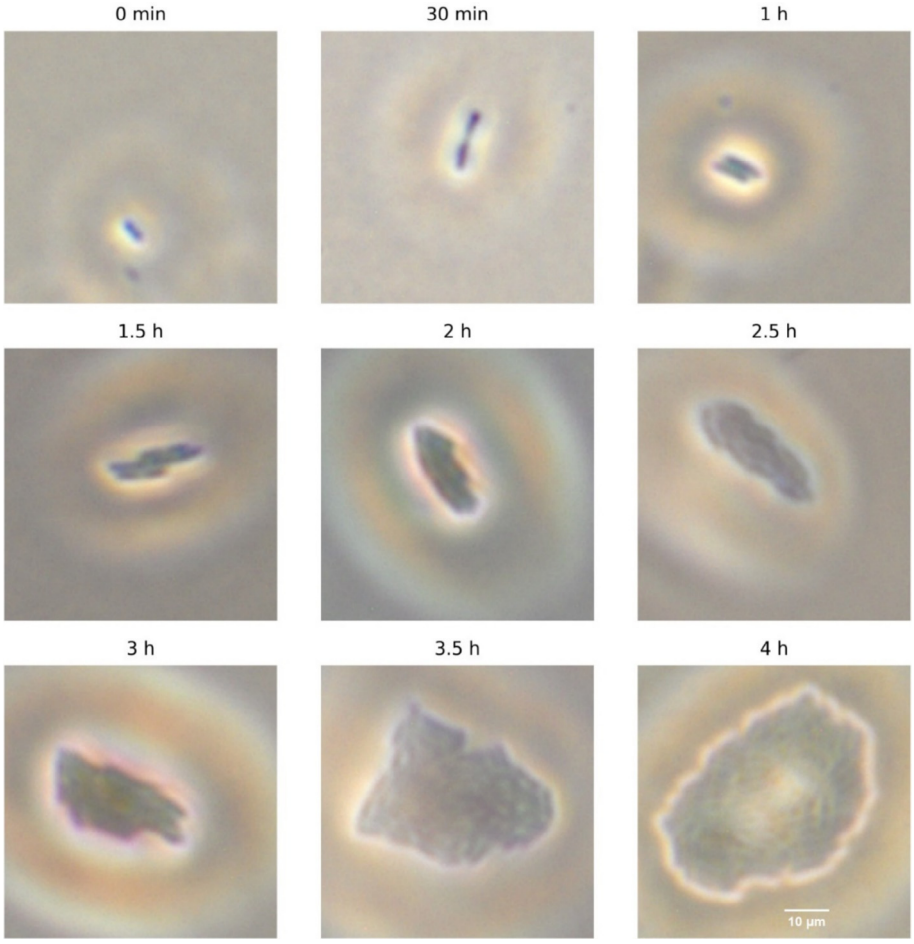


Figure 1. Sequential microscopic images showing the growth progression of *S.* Typhimurium from 0 min to 4 h of incubation.

tion from monolayer to multilayer organization was observed after approximately 3.5 h of incubation, indicating increased structural complexity during later growth stages (You et al., 2018).

Quantitative analysis demonstrated a gradual increase in mean microcolony area during the first 3 h, followed by a phase of accelerated expansion (Fig. 2). The mean colony area reached 48 μm^2 at 2 h and increased more than 40-fold by 4 h of incubation. Notably, by 2 h, microcolonies had developed sufficient biomass and morphological features to enable reliable downstream identification, and this time point was therefore selected for early detection.

YOLOv8x detection performance. The performance of the YOLOv8x model for microcolony detection and identification was further evaluated across multiple time points. Early-stage images (≤1.5 h) presented challenges due to limited morphological differentiation; therefore, subsequent analyses focused on later time points where microcolonies were more distinct.

High annotation quality was achieved during dataset preparation, with an interannotator agreement of 97.3%, reflecting strong consistency in microcolony labeling (Fig. 3). The YOLOv8x model integrates detection and identification into a single inference step (Tachaanantakit & Rattanatamrong, 2025), enabling efficient and real-time identification of both *S.* Typhimurium and *E. coli* K-12 microcolonies separately (Fig. 4).

Quantitative evaluation demonstrated robust model performance, with a precision of 0.891, a recall of 0.867, and a mean average precision at 0.5 IoU (mAP@50) of 0.898. These results are consistent with those reported in our previous study (AlSobeh et al., 2025). Qualitative assessment further confirmed accurate localization of microcolonies using well-defined bounding boxes and correct class assignment in most cases. The model maintained consistent performance across mixed cultures and multiple imaging time points, indicating strong generalization under the tested conditions.

Identification of bacterial microcolonies using YOLOv8x. To evaluate the ability of the model to distinguish between bacterial species, *E. coli* K-12 microcolonies were imaged under identical conditions (Fig. 5). Mixed cultures containing *S.* Typhimurium and *E. coli* K-12 at an 8:2 volumetric ratio were used to assess identification performance.

The YOLOv8x model demonstrated strong discriminatory capability, as evidenced by the confusion matrix (Fig. 6). High correct identification rates were achieved for both *S.* Typhimurium (0.89) and *E. coli* K-12 (0.88), with minimal interclass misidentification. Specifically, 0.06 of *E. coli* K-12 instances were incorrectly identified as *S.* Typhimurium, while 0.04 of *S.* Typhimurium instances were misidentified as *E. coli K-12*. The confusion matrix is column-normalized following the Ultralytics convention. Background is not an enumerated class in object detection, and no background-background entry therefore exists; the background column contains only predictions that were not matched to a ground-truth instance, normalized over that set alone. The values in this column (0.58 and 0.42) describe how unmatched detections were distributed between the two bacterial classes and do not represent a false-positive rate. The corresponding absolute performance is given by the overall precision of 0.89 reported. The bottom row indicates that 0.05 of *S.* Typhimurium and 0.08 of *E. coli* K-12 instances were not detected.

Overall, these results demonstrate that YOLOv8x enables reliable discrimination between morphologically similar bacterial microcolonies in mixed cultures (Fig. 7).

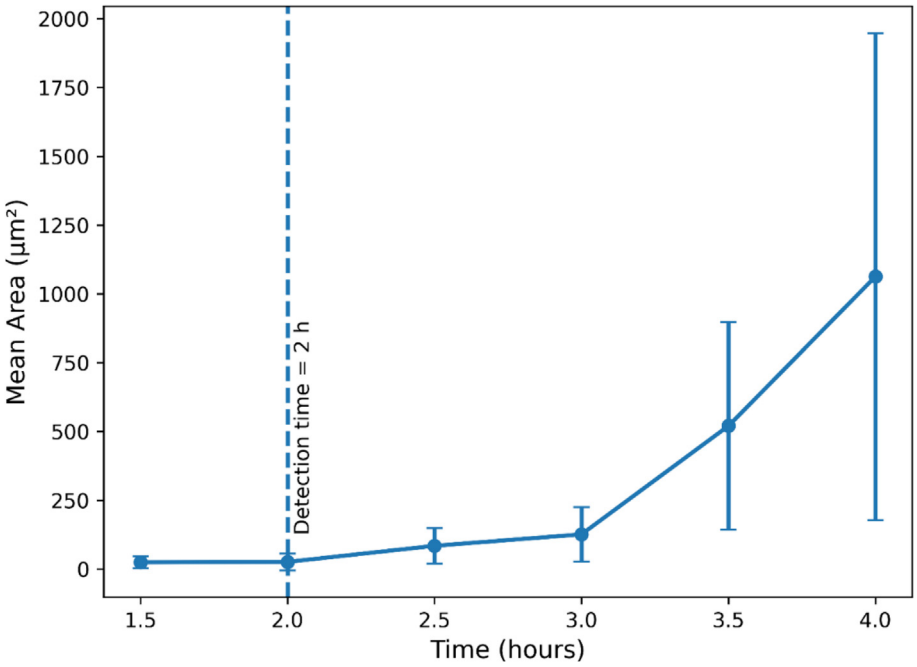


Figure 2. Early microcolony growth dynamics of *S.* Typhimurium during incubation. The error bars indicate the standard deviation.

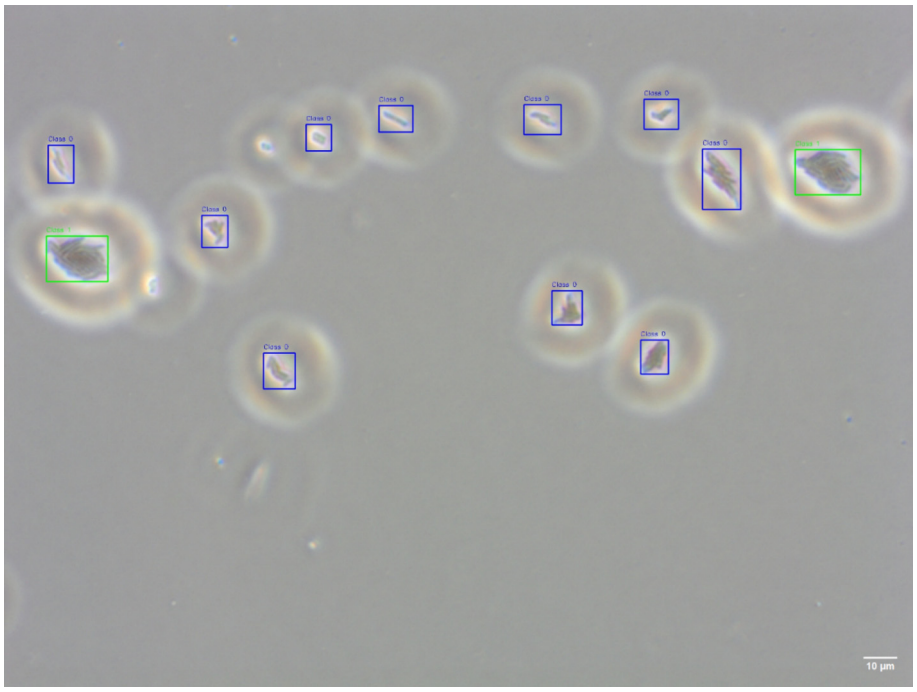


Figure 3. Polygon-based annotation of microcolonies.

Detection under low-level contamination conditions. To further evaluate the performance of the proposed framework under low-level contamination conditions, additional experiments were conducted using serially diluted *S.* Typhimurium inocula in onion samples. Two low contamination levels, 100 CFU/g and 10 CFU/g, were evaluated using 10 samples at each concentration. At 100 CFU/g, *S.* Typhimurium microcolonies were successfully identified in 7 of 10 samples (70%) using phase-contrast microscopy and the YOLOv8x-based detection framework. When the contamination level was further reduced to 10 CFU/g, microcolonies were not detected in any of the 10 samples evaluated under the experimental conditions. These results demonstrate successful detection at 100 CFU/g while indicating reduced detection capability at the lower contamination level of 10 CFU/g.

Detection under low-level contamination conditions. To further evaluate the performance of the proposed framework under low-level contamination conditions, additional experiments were conducted using serially diluted *S.* Typhimurium inocula in onion samples. Two low contamination levels, 100 CFU/g and 10 CFU/g, were evaluated using 10 samples at each concentration. At 100 CFU/g, *S.*

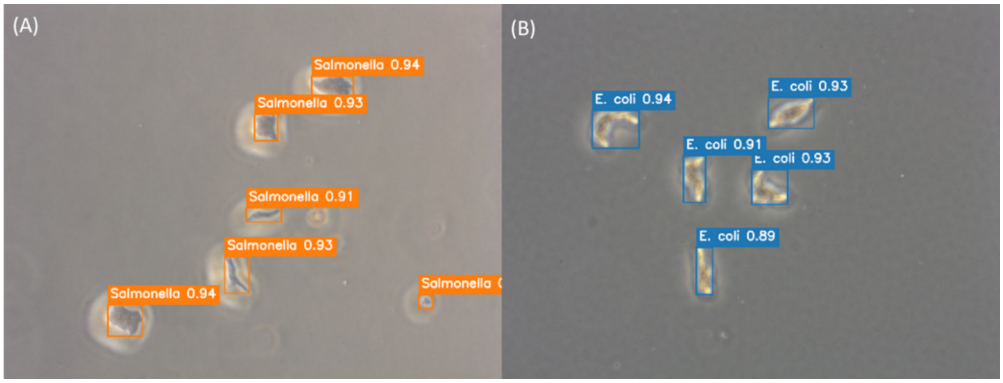


Figure 4. Microcolony detection using YOLOv8x model: (A) detection results for *S.* Typhimurium; (B) detection results for *E. coli* K-12.

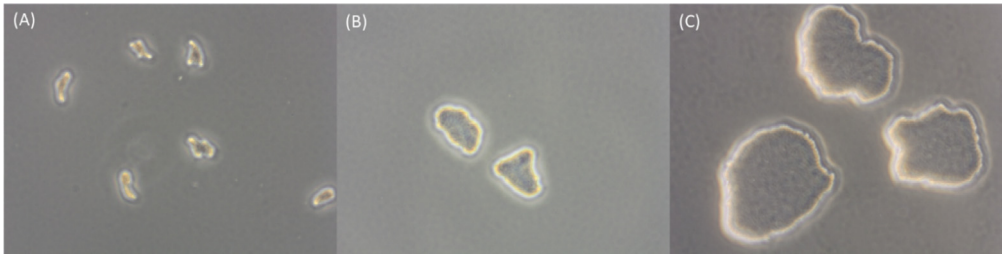


Figure 5. Representative image of *E. coli* K-12 microcolonies at (A) 2 h of incubation, (B) 3 h of incubation, (C) 4 h of incubation.

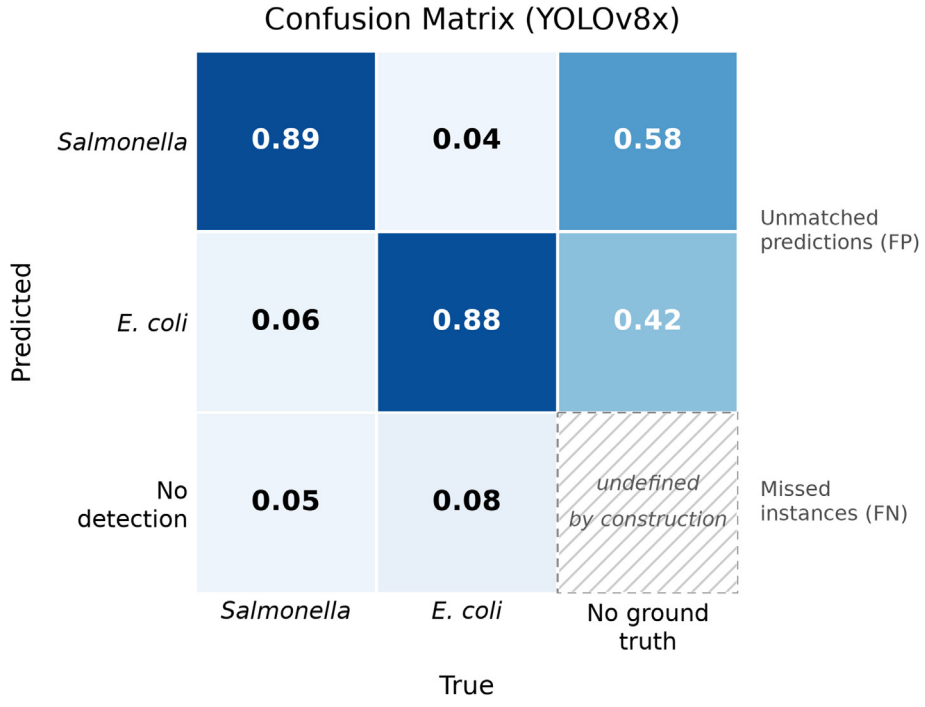


Figure 6. Confusion matrix for microcolony detection and identification by the YOLOv8x model, pooled across imaging time points (2–4 h), single and mixed cultures, and both plain and onion-matrix conditions. Values are column-normalized following the Ultralytics convention. The third column ("no ground truth") contains only predictions that were not matched to an annotated instance, normalized over that set alone; its values therefore describe how unmatched detections were distributed between the two bacterial classes and do not represent a false-positive rate.

Typhimurium microcolonies were successfully identified in 7 of 10 samples (70%) using phase-contrast microscopy and the YOLOv8x-based detection framework. When the contamination level was further reduced to 10 CFU/g, microcolonies were not detected in any of the 10 samples evaluated under the experimental conditions. These results demonstrate successful detection at 100 CFU/g while indicating reduced detection capability at the lower contamination level of 10 CFU/g.

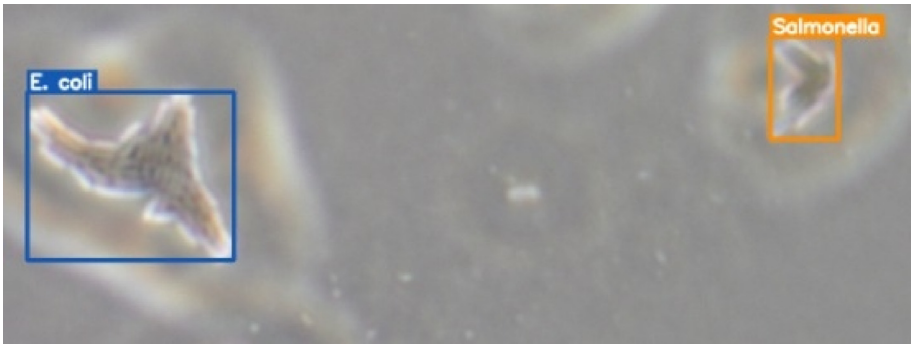


Figure 7. Microcolony detection results for *S.* Typhimurium and *E. coli* K-12 in mixed culture.

Quantitative analysis of microcolony size using instance segmentation. Quantitative analysis of *S.* Typhimurium microcolony size revealed a time-dependent increase in colony area. At 2 h, the average microcolony size was 48 μm^2, increasing to 105 μm^2 at 2.5 h and 125 μm^2 at 3 h. A pronounced expansion was observed at 4 h, where the average size reached 930 μm^2, indicating a transition to rapid growth.

Representative phase-contrast images and corresponding instance segmentation masks further illustrate the progressive increase in both cell number and colony area over time (Figs. 8 and 9).

Discussion

In this study, we demonstrate that a YOLOv8x-based imaging framework, integrated with phase-contrast microscopy, enables reliable detection and identification of *S.* Typhimurium at the microcolony stage within a 2-h incubation period. Our results show that distinct microcolonies emerge between 2 and 3 h and exhibit rapid expansion thereafter, with a marked increase in colony size by 4 h. Importantly, quantitative analysis revealed that microcolonies reached an average size of 48 μm^2 at 2 h, providing sufficient morphological features for accurate identification. This finding demonstrates the feasibility of recognizing *S.* Typhimurium microcolonies at an early stage of growth under controlled laboratory conditions, significantly reducing the time required compared with conventional culture-based methods.

Traditional methods for *Salmonella* detection, although sensitive and widely adopted, are time-intensive and labor-dependent, typically requiring several days for visible colony formation and confirmatory testing (Andrews & Ryan, 2015). Advanced immunological and molecular approaches allow for rapid detection within a few hours while maintaining high sensitivity and specificity. However, immunological methods can be limited by antibody instability and cross-reactivity, whereas molecular techniques such as PCR, LAMP, and RPA often require specialized instrumentation, may be influenced by inhibitory substances in samples, and are unable to differentiate between viable and nonviable cells, thereby restricting their effectiveness in real-world applications (Silva et al., 2018; Sue et al., 2014; Li et al., 2017; Rosser et al., 2015). In contrast, the approach presented here provides a rapid, automated, and cost-effective alternative, with the potential for high-throughput screening and on-site application.

The observed microcolony growth dynamics further support the suitability of early detection. Time-resolved imaging revealed a transition from single cells to structured microcolonies, followed by a shift from monolayer to multilayer organization after approximately 3.5 h. This progression is consistent with known bacterial growth behavior and underscores the importance of capturing morphological changes over time. By leveraging these dynamic features, our method enhances discriminatory capability beyond what is achievable with static endpoint measurements.

The YOLOv8x model demonstrated strong performance in both detection and identification tasks, achieving a precision of 0.891, a recall of 0.867, and mAP@50 of 0.898. The confusion matrix analysis further confirmed high true positive rates for both *S.* Typhimurium and *E. coli*, with minimal interclass misidentification. A proportion of detections was not matched to annotated instances and is reported in the background column of the confusion matrix; these unmatched predictions are consistent with the presence of visually similar non-colony features, including onion-derived matrix debris, in phase-contrast images of this food matrix. Nonetheless, the overall perfor-

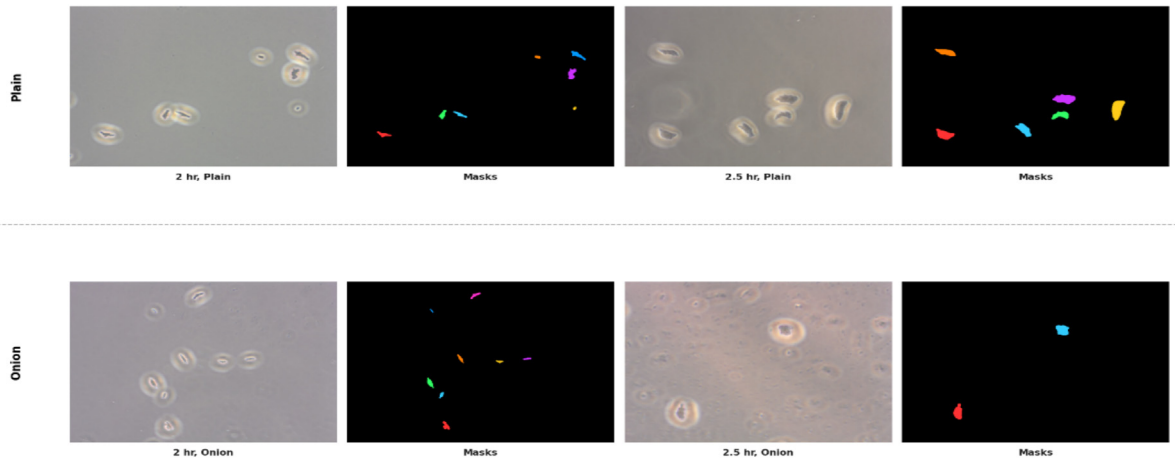


Figure 8. Representative phase-contrast images and corresponding instance segmentation masks of *S.* Typhimurium microcolonies at early time points (2–2.5 h) under different treatment conditions.

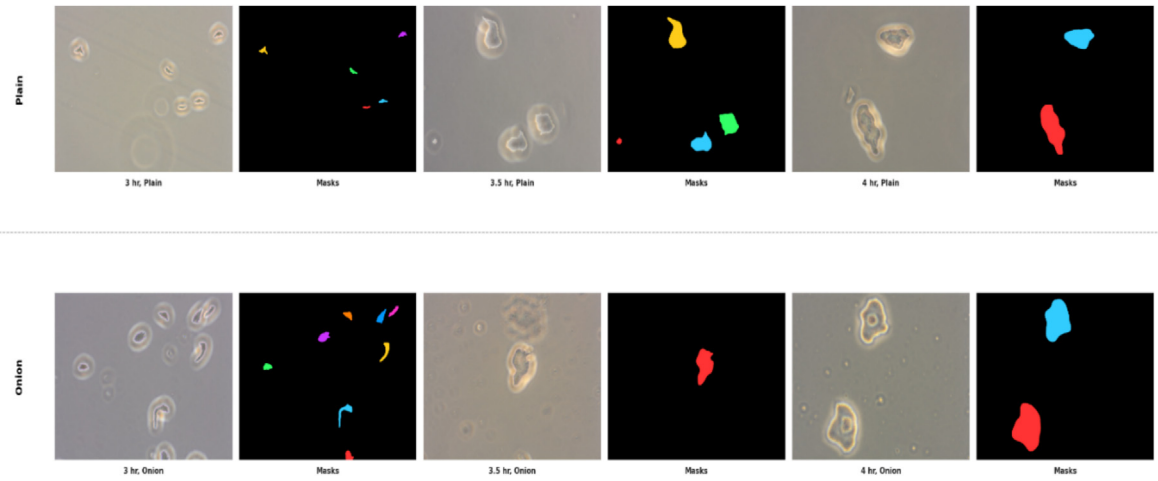


Figure 9. Representative phase-contrast images and corresponding instance segmentation masks showing microcolony development of *S.* Typhimurium at later time points (3–4 h), highlighting increased colony size and cell number.

mance indicates that YOLOv8x is a robust tool for real-time bacterial identification in mixed cultures.

Compared with previous studies, the present approach demonstrates clear advantages in both dataset scale and detection speed. Earlier methods based on colony fingerprinting (CFP) and conventional machine learning required extended incubation periods (up to 20 h) and were limited by relatively small datasets (Tanaka et al., 2019). Recent advances in deep learning have enabled rapid and accurate bacterial detection; for example, AI-enabled lab-on-chip systems have reported detection accuracies of up to 95% with real-time analysis, thereby substantially reducing turnaround times compared with conventional techniques (Nehal et al., 2020). Nevertheless, these systems are typically evaluated under controlled experimental conditions with limited dataset variability, which may constrain their applicability in complex, real-world environments. Similarly, prior deep learning studies employing YOLO-based architectures have reported promising results; however, their performance has often been constrained by limited dataset size and variability. For example, Ma et al. (2023) reported high detection accuracy using YOLOv4, but their model was trained on only 315 images per bacterial strain, potentially limiting generalizability. In contrast, the current study utilizes a larger dataset, contributing to enhanced robustness and rapid detection capability.

Furthermore, the DeepBacs framework highlights that the performance of deep learning models in bacterial image analysis is strongly dependent on the quality, diversity, and scale of the training data, underscoring the importance of well-curated datasets for reliable and reproducible results. Deep learning approaches have also demonstrated high accuracy and rapid processing across multiple tasks, including segmentation, object detection, and identification, supporting their advantages over conventional methodologies (Spahn et al., 2022). In contrast, the present study employs a larger dataset, which contributes to enhanced model robustness and improved detection efficiency.

The integration of instance-level segmentation further enabled quantitative characterization of microcolony growth. The observed increase in colony size, from 48 μm^2 at 2 h to 930 μm^2 at 4 h, demonstrates a transition from early-stage formation to rapid expansion. This capability extends the utility of the method beyond detection to include functional assessment of bacterial behavior under different conditions.

From an application perspective, the ability to detect *S.* Typhimurium within 2 h has significant implications for food safety monitoring. Rapid identification at the microcolony stage can enable earlier intervention in contamination events, reducing the risk of widespread outbreaks. Furthermore, the combination of automated image analysis and real-time detection supports scalability and integration into routine screening workflows. This approach has the potential to support rapid screening in food processing environments and reduce the time to intervention during contamination events.

Despite these promising findings, several limitations should be considered. First, this study was performed under controlled laboratory inoculation conditions using experimentally inoculated samples to facilitate systematic image acquisition and deep learning model development. Consequently, the proposed framework can detect S. Typhimurium at a low contamination level of 100 CFU/g, while also demonstrating that detection becomes more challenging as the initial contamination level decreases. At very low contamination levels, detection may be influenced by the heterogeneous distribution of bacterial cells within the food matrix and the probability that developing microcolonies are present within the microscopic field of view. Therefore, 100 CFU/g is reported as the lowest contamination level at which successful detection was demonstrated in this study. Second, only a limited number of bacterial species and one food matrix were evaluated. Future studies should assess the complete workflow using naturally contaminated samples, lower inoculum concentrations, diverse food matrices, and additional bacterial species to determine practical detection limits and operational performance under real-world conditions. Third, model selection was based on architectural rationale and prior application to this imaging modality rather than on an empirical head-to-head comparison; a controlled benchmark against alternative detectors, including more recent YOLO variants and two-stage architectures such as Faster R-CNN, evaluated specifically on small-object performance in low-contrast phase-contrast microscopy and stratified by microcolony size, is an important direction for subsequent work. Fourth, robustness against matrix-derived debris would be further strengthened by expanded annotation of visually challenging non-colony features and by hard-negative mining during training.

Conclusion

This study demonstrates that the integration of YOLOv8x-based object detection with time-resolved phase-contrast imaging provides an effective and rapid framework for early detection and identification of *S.* Typhimurium. By significantly reducing detection time while maintaining high accuracy, this approach represents a promising advancement in image-based microbial diagnostics and food safety applications. Future work will focus on expanding the model to diverse food matrices and microbial communities to further enhance its practical applicability.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT (OpenAI) to assist with language editing and improving clarity. After using this tool, the author(s) carefully reviewed and edited the content as needed and took full responsibility for the content of the published article.

CRediT authorship contribution statement

Namariq Dhahir: Writing – original draft, Validation, Resources, Methodology, Investigation, Data curation, Conceptualization. **Malek Rababa:** Writing – review & editing, Software, Methodology, Investigation, Data curation. **Bibek Koirala:** Writing – review & editing, Software, Methodology, Formal analysis, Data curation. **Amer AbuGhazaleh:** Writing – review & editing, Validation, Project administration, Funding acquisition. **Anas AlSobeh:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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